



Faculty of Informatics & Design

Department of Information Technology

PRJ470S – Advanced Diploma in Information Technology

Assessment Two: Research Proposal Submission

TITLE OF RESEARCH PROPOSAL

**The Adoption of Cloud Computing by Small to Medium
Enterprises in South Africa**

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Qualification: Projects 4: Advanced Diploma in ICT (Applications Development)

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Plagiarism Clause and AI Declaration

I, Thabiso Patrick Matsaba, hereby declare that the research proposal submitted herewith is my own original work. I acknowledge that I am fully aware of the Cape Peninsula University of Technology (CPUT) Policy on Plagiarism and Academic Dishonesty, and I confirm that all literary and empirical sources utilized in this document have been explicitly and systematically acknowledged through parenthetical in-text citations and a complete reference list compiled in strict accordance with the prescribed CPUT Harvard Referencing Style.

I further declare that this document is free from plagiarism, and its primary conceptual framework, theoretical synthesis, research sub-questions, and systematic methodology have been originally conceived and written by myself. No part of this proposal has been previously submitted for an academic qualification at this or any other tertiary institution.

Ethical Disclosure of Generative AI Tool Usage:

In alignment with the CPUT Student Practice Guide for Generative AI and the Projects 4 Assessment Rubric, the precise nature of generative artificial intelligence integration in compiling this proposal is documented below. The use of the tool was strictly confined to non-cognitive, editorial, and linguistic functions, ensuring that scholarly ownership and cognitive synthesis remain strictly with the researcher.

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Was generative AI utilized during the drafting of this research proposal?	[X] YES / [] NO Tool utilized: Gemini NotebookLLM.
Specify the exact scope and operational boundaries of the AI tool's application.	The generative AI tool was utilized strictly as an editorial assistant. Its application was confined to: (1) optimizing sentence flow and linguistic transitions, (2) verifying spelling, grammar, and syntax, and (3) ensuring formatting compliance with CPUT style manuals (Calibri 11pt, 1.15-line spacing, structured headings).
Confirm ownership of the primary research ideas, methodology, and empirical analysis.	The candidate confirms that the core research concepts, the integration of the Technology-Organization-Environment (TOE) framework, the formulation of research questions and objectives, the systematic database search parameters, the selection of inclusion/exclusion criteria, and the references are the original intellectual work of the candidate.

Candidate Signature: _____

Date: September 2026

1. Introduction

In recent years, technology has transformed the approaches and directions in which many businesses operate in this era of technological advancement. The widespread move towards cloud computing represents one of the most significant developments in this transformation. Cloud computing offers organizations the affordability to run software applications, manage business systems, and store and retrieve information enabled by internet connectivity, without the need to invest in expensive physical infrastructure (Jayeola et al., 2022:228; Mokwena & Sithole, 2024:12; Mkhize et al., 2025:5). According to the National Institute of Standards and Technology (NIST), cloud computing is defined as a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction (Mokwena & Sithole, 2024:12; Mell & Grance, 2011:SP 800-145). This model represents a fundamental departure from on-premises computing, shifting IT investment from capital expenditure (CapEx) to operational expenditure (OpEx), which is highly appealing for businesses operating with tight capital constraints (Jayeola et al., 2022:230; McKinsey & Company, 2024:2).

Small and Medium Enterprises (SMEs) serve as the backbone of the South African economy. Both formal and informal South African SMEs account for more than 90 percent of all businesses, employing approximately 60 percent of the national workforce and contributing around 40 percent of the Gross Domestic Product (Ngo Ndjama & Van der Westhuizen, 2024:606; DSBD, 2023:12). The National Development Plan (NDP 2030) further projects that SMEs should create 90 percent of the eleven million jobs the government seeks to achieve by 2030 (South Africa. Department of Small Business Development [DSBD], 2023:12). Given their pivotal economic role, the need for SMEs to operate more effectively and competitively has never been more urgent. The adoption of cloud computing can provide SMEs with a platform for scalability, enabling them to access enterprise-grade tools, reduce operational overhead, improve business agility, and respond to rapid market changes without the typical constraints of traditional, static IT infrastructure (Ilahi et al., 2024:57; Mkhize et al., 2025:6).

Despite the well-documented potential of cloud computing to reduce costs and improve operational efficiency, South African SMEs continue to adopt cloud-based services at a pace that is far below their potential (Mkhize et al., 2025:8). Research conducted across the African continent consistently reveals that SMEs face a complex, interlocking combination of technological, organizational, and environmental barriers that hinder their digital transformation (Jayeola et al., 2022:232; IJSRA, 2024:1445). These barriers are highly context-sensitive and unequal: an SME located in an urban hub like Johannesburg may struggle primarily with high data costs and complex Protection of Personal Information Act (POPIA) data privacy compliance, while an enterprise in a rural province like Limpopo must additionally contend with persistent utility infrastructure failures, such as load shedding, and a severe localized digital skills deficit (Mokwena & Sithole, 2024:14).

Understanding why cloud adoption remains critically low requires a robust, structured theoretical lens that spans more than just technical features. This study employs the Technology-Organisation-Environment (TOE) framework, originally developed by Tornatzky and Fleischer (1990) and widely validated in information systems research, as its primary analytical tool to identify and categorize the factors influencing cloud computing adoption decisions among South African SMEs (Adane, 2021:4; Santos et al., 2023:Article 2220190). The TOE framework organizes adoption influences into three interacting contexts: technological, organisational, and environmental (Tornatzky & Fleischer, 1990:152). This makes it exceptionally well-suited to evaluating the multi-dimensional socio-technical barriers facing SMEs in developing economies, where infrastructural gaps and shifting regulatory mandates directly impact business viability (Abu Bakar et al., 2024:556; Ilahi et al., 2024:58). Ultimately, this study seeks to propose a contextualized cloud adoption framework that can guide South African SMEs towards sustainable, compliant, and cost-effective digital transitions.

2. Background to the Research Problem

South Africa possesses one of the most advanced ICT infrastructures on the African continent, yet a severe digital divide persists between how large, resource-rich corporations leverage technology and how small businesses operate. While major enterprises have the financial and human capital to adopt complex cloud computing architectures in full, the vast majority of local SMEs still function with basic on-premises digital tools, limited internet connectivity, and a lack of baseline knowledge regarding cloud capabilities (IJSRA, 2024:1446). This digital exclusion reflects deep-seated, systemic inequalities in access to capital, technical skills, and reliable national infrastructure that have persisted across the post-apartheid era (FinMark Trust, 2024:3).

SMEs are formally defined under the South African National Small Business Act (Act 102 of 1996, as amended) as separately owned and managed entities that fall within specific thresholds based on full-time employee count and annual turnover (DSBD, 2023:14). In most South African sectors, a micro-enterprise employs fewer

than five employees, a small enterprise between six and fifty, and a medium enterprise between fifty-one and two hundred employees (DSBD, 2023:14). Despite being the primary engine of employment, accounting for 55% of the total workforce, these businesses experience an exceptionally high failure rate, with a 15.9% year-on-year decrease in job provision recorded during economic downturns (Sithole & Ruhode, 2021:283; SEDA, 2018:4). This high failure rate is heavily compounded by an inability to modernize business operations due to resource constraints and information technology deficiencies (Mokwena & Sithole, 2024:18; Fatoki, 2014:922).

Furthermore, migrating to the cloud introduces distinct regulatory and infrastructural challenges that distinguish South Africa from global contexts. The Protection of Personal Information Act (POPIA), which came into full regulatory enforcement in July 2021, enforces strict mandates on the cross-border transfer of personal data (Mokwena & Sithole, 2024:16; Mohlameane & Ruxwana, 2020:4). Because many global cloud hyperscalers store data in international data centers, South African SMEs face severe compliance anxieties. Lacking in-house legal expertise or specialized IT staff, these businesses frequently choose to avoid cloud migration entirely to prevent potential regulatory penalties (Mokwena & Sithole, 2024:22). Compounding this is the national energy crisis, where persistent load shedding disrupts local internet routers and cellular towers, making cloud-hosted systems temporarily inaccessible and exposing SMEs to operational vulnerabilities that are treated as non-existent in global literature (Adane, 2021:6; Mokwena & Sithole, 2024:28).

2.1 Research Problem

South African SMEs are currently facing a critical lag in digital transformation, as their cloud computing adoption rates remain disproportionately low compared to the global average. This digital stagnation prevents small businesses from operating competitively, directly leading to high rates of business failure and undermining national employment and economic targets. Empirical research shows that typical South African SMEs remain heavily reliant on obsolete on-premises IT methods due to severe constraints, such as limited technical literacy, initial capital barriers, data security anxieties, regulatory compliance hurdles, and persistent utility failures (Mokwena & Sithole, 2024:18; Mkhize et al., 2025:14). If these multi-dimensional technological, organisational, and environmental barriers are not systematically addressed through an integrated theoretical framework, South African SMEs will remain digitally excluded, fail to survive under intense competitive pressure from larger enterprises, and be unable to support the National Development Plan (NDP) target of generating 90% of new employment by 2030 (DSBD, 2023:15; Tornatzky & Fleischer, 1990:152).

Research Problem Statement: *South African small and medium enterprises are not adopting cloud computing configurations at a rate commensurate with its potential to improve their operations, and the combined technological, organisational, and environmental factors inhibiting this transition have not been comprehensively examined through an integrated, locally-relevant theoretical framework.*

2.2 Aim of the Study

The primary aim of this study is to investigate the technological, organisational, and environmental factors that influence cloud computing adoption decisions among small and medium enterprises in South Africa, utilizing the TOE framework as an analytical lens, with a view to proposing a contextualised adoption framework that can systematically guide local business owners, policymakers, and cloud vendors in accelerating secure, compliant, and cost-effective cloud transitions.

2.3 Objectives of the Study

In pursuit of the above aim, this study is guided by the following three objectives:

1. To identify the technological factors that hinder or enable cloud computing adoption among South African SMEs.
2. To examine the organisational and environmental factors that influence cloud computing adoption decisions among South African SMEs.
3. To propose a framework that can guide South African SMEs in making informed and sustainable cloud computing adoption decisions aligned with their specific technological, organisational, and environmental contexts.

2.4 Main Research Question

What are the key technological, organisational, and environmental factors that influence cloud computing adoption among small and medium enterprises in South Africa, and how can these factors be addressed through a contextualised adoption framework?

2.5 Sub-Questions

To operationalise the main research question, the study is further guided by the following sub-questions, each aligned with a specific construct of the TOE framework:

1. What are the technological factors that hinder or enable cloud computing adoption among South African SMEs?
2. What organisational factors influence the cloud computing adoption decisions of South African SMEs?
3. What environmental factors shape SME decisions to adopt or avoid cloud-based systems in South Africa?
4. How can a framework, informed by the TOE lens, be structured to guide South African SMEs in adopting cloud computing in a manner that is affordable, legally compliant, and operationally appropriate for their specific context?

3. Literature Review

This literature review critically evaluates the body of academic knowledge concerning the adoption of cloud computing configurations by small and medium enterprises, with a specific focus on the unique regulatory, socio-technical, and economic conditions of South Africa. Rather than merely describing individual studies in a descriptive manner, this review adopts a critical synthesis approach, mapping patterns, contradictions, and empirical gaps across national and global literature. The synthesis is structured around the three-dimensional Technology-Organisation-Environment (TOE) framework proposed by Tornatzky and Fleischer (1990:152), which provides a robust and validated lens for assessing innovation adoption in resource-constrained organisational settings (Abu Bakar et al., 2024:556; Santos et al., 2023:Article 2220190).

3.1 Cloud Computing

To evaluate cloud adoption decisions, it is technically necessary to define and dissect the computing configurations available on the modern market. According to the National Institute of Standards and Technology (NIST), cloud computing consists of five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service (Mokwena & Sithole, 2024:12; Mell & Grance, 2011:SP 800-145). These characteristics collectively distinguish cloud models from traditional computing. Under an on-demand model, a business can provision computing capabilities automatically without human intervention from the service provider, while broad network access ensures capabilities are available over the network and accessed through standard platforms (such as mobile phones, laptops, and tablets) (Mokwena & Sithole, 2024:12; Mokwena & Fatoki, 2024:4627).

Cloud capabilities are structured and delivered through three primary service models, each representing different levels of administrative control and client responsibility (Ahmad et al., 2025:1; Greiner, 2014:2). These are Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), and Infrastructure-as-a-Service (IaaS). Under the Software-as-a-Service (SaaS) model, the customer is provided access to the provider's applications running on a cloud infrastructure, which are accessible through a thin client interface such as a web browser (e.g., Gmail, Google Maps, Dropbox, and cloud-based Point-of-Sale systems) (Mokwena & Fatoki, 2024:4627; Mokwena & Sithole, 2024:284). Under SaaS, the SME does not manage or control the underlying cloud infrastructure, including servers, operating systems, storage, or individual application capabilities, making it the most affordable, user-friendly, and popular choice for resource-constrained small businesses with limited IT skills (Ongowarsito et al., 2022:24; Li & Kumar, 2022:2588).

Conversely, Platform-as-a-Service (PaaS) provides the client with a structured environment to deploy onto the cloud infrastructure consumer-created or acquired applications built using programming languages, libraries, services, and tools supported by the provider (e.g., Google App Engine, Microsoft Azure DevOps, AWS Elastic Beanstalk) (Mokwena & Fatoki, 2024:4627; Wen et al., 2024:2240). The client gains control over the deployed applications and possibly configuration settings for the application-hosting environment, while the underlying infrastructure is entirely managed by the cloud service provider (CSP). This model is highly valuable for SMEs engaged in custom software development, enabling rapid testing and deployment without hardware management overhead (Bharany et al., 2022:5013). Finally, Infrastructure-as-a-Service (IaaS) delivers fundamental computing resources, including physical or virtual machines, raw storage, firewalls, and networks, on an on-demand, pay-as-you-go basis (e.g., Amazon Web Services [AWS] EC2, Microsoft Azure Virtual Machines) (Homan & Beránek, 2023:296; Saraswat & Tripathi, 2020:300). While IaaS offers the highest level of administrative control and flexibility, it places the sole responsibility for operating systems, middleware, database engines, and applications upon the SME, demanding advanced internal IT expertise that typical small businesses completely lack (Kondo et al., 2022:253; Mkhize et al., 2025:14).

Cloud configurations are further defined by four distinct deployment models: public, private, community, and hybrid (Mokwena & Fatoki, 2024:4627). Public clouds are owned and operated by third-party CSPs and shared across multiple organizations, representing the most cost-effective and scalable configuration but offering minimal control over data residency. Private clouds are provisioned for exclusive use by a single organization, providing robust security and control but requiring massive capital investment that violates the budget

constraints of most SMEs. Community clouds are shared among organizations with common concerns (such as industry-specific compliance standards). Hybrid clouds represent a composition of two or more distinct cloud infrastructures (public, private, or on-premises) that remain bound together by standardized or proprietary technology that enables data and application portability (Mokwena & Fatoki, 2024:4627). Hybrid models are increasingly recognized as the most pragmatic configuration for South African SMEs, enabling them to leverage the cost-efficiency of public clouds for non-sensitive data while keeping sensitive customer data secure and legally compliant on private or on-premises servers (McKinsey & Company, 2024:2; Neves et al., 2011:5).

3.2 The Technology-Organisation-Environment (TOE) Framework

First conceptualized by Tornatzky and Fleischer (1990:152), the TOE framework serves as the primary theoretical model for this study. The model proposes that the adoption and implementation of any technological innovation is shaped by three distinct, interacting contexts within an enterprise: technological, organisational, and environmental contexts (Tornatzky & Fleischer, 1990:152). Unlike simpler, user-centric adoption models like the Technology Acceptance Model (TAM), which evaluate individual perceptions of usefulness and ease of use (Davis, 1989:319), the TOE framework operates at the firm level, enabling a holistic analysis that integrates technical capabilities with organisational constraints and external institutional forces (Jayeola et al., 2022:235; Santos et al., 2023:Article 2220190). Consequently, the TOE framework is highly appropriate for examining cloud adoption barriers in developing economies, where utility instability and complex regulatory mandates (such as POPIA) represent critical factors that cannot be analyzed through purely technical lenses (Abu Bakar et al., 2024:556; Mokwena & Sithole, 2024:30).

3.3 Technological Factors Influencing Cloud Adoption

Within the technological context, relative advantage represents the primary driver of cloud adoption, defined as the degree to which an innovation is perceived as superior to existing traditional infrastructure (Tornatzky & Fleischer, 1990:152). Global empirical studies consistently demonstrate that when SME managers recognize that cloud configurations lower IT costs, improve operational efficiency, and enhance team collaboration, their adoption intention rises significantly (Ilahi et al., 2024:57; Santos et al., 2023:Article 2220190). However, a sharp contradiction emerges in the South African literature. An empirical study of 230 SMEs in the Polokwane Municipality found that local business owners frequently do not perceive any relative advantage in cloud migration, viewing it as an unnecessary, complex expense (Mokwena & Sithole, 2024:20). This conflict indicates that relative advantage is not an inherent feature of cloud technology but depends heavily on prior digital literacy and IT awareness among SME executives (Mokwena & Fatoki, 2024:261).

Furthermore, compatibility and complexity serve as major technological determinants. Compatibility assesses how seamlessly cloud solutions integrate with an enterprise's existing business workflows, culture, and legacy systems (Yaseen et al., 2023:320). If an SME relies on highly customized, on-premises legacy databases that cannot connect to modern cloud environments, the resultant incompatibility leads to severe cost inflation and project failure (Mkhize et al., 2025:12). Complexity, representing the perceived difficulty of configuring and managing cloud services, acts as a primary barrier (Yaseen et al., 2023:320). While simple SaaS products present low complexity, configuring IaaS and PaaS environments demands advanced IT capabilities. Lacking internal IT personnel, South African SMEs frequently reject cloud adoption due to fear of technical complexity (Mokwena & Fatoki, 2024:261; Mkhize et al., 2025:14). Data security and POPIA compliance represent the final, critical technological barrier in South Africa. Unlike developed economies where security is treated as a manageable technical factor, South African SME owners view data security as an absolute barrier due to deep mistrust of remote, third-party storage (Jayeola et al., 2022:228). This is heavily exacerbated by POPIA (Act 4 of 2013), which came into full enforcement in July 2021 and mandates strict penalties for non-compliant cross-border transfers of personal data (Mokwena & Sithole, 2024:16; Mohlameane & Ruxwana, 2020:4). Because global cloud vendors use international data centers, local SMEs without specialized legal counsel choose to remain on obsolete on-premises systems to avoid regulatory violations (Mokwena & Sithole, 2024:22).

3.4 Organisational Factors Influencing Cloud Adoption

Within the organisational context, top management support represents the most critical determinant of adoption success (Tornatzky & Fleischer, 1990:152). In small enterprises characterized by highly centralized, flat managerial structures, the personal attitude and digital literacy of the business owner dictate the adoption trajectory (Abu Bakar et al., 2024:556). If the owner actively supports digital transformation, they can successfully allocate necessary resources and overcome internal resistance to change. However, if leadership is risk-averse, lacks basic IT awareness, or views cloud services merely as a financial expenditure, the enterprise remains locked in obsolete operational models (Abu Bakar et al., 2024:556; Mokwena & Fatoki, 2024:261). Crucially, financial resource readiness acts as a primary filter. While CSPs market the cloud as a cost-saving tool, the initial migration costs—including internet bandwidth upgrades, data migration consultant

fees, dual-system subscription costs, and staff training—represent an insurmountable barrier for resource-constrained South African SMEs (McKinsey & Company, 2024:4; Jayeola et al., 2022:232).

The foremost organisational barrier in South Africa is the acute digital skills shortage and absence of dedicated IT personnel (Mkhize et al., 2025:14; Reviewing the Impact, 2024:1445). Large enterprises maintain internal IT departments to manage complex systems, but typical South African SMEs are managed by self-taught owners who completely lack formal technical training (Mkhize et al., 2025:14). Lacking internal technical expertise, these businesses are unable to evaluate SLA contracts, manage data migration pipelines, or troubleshoot connectivity issues, thereby escalating their reliance on expensive external consultants and severely inflating the total cost of ownership (Mokwena & Fatoki, 2024:280; Mkhize et al., 2025:14).

3.5 Environmental Factors Influencing Cloud Adoption

Within the environmental context, competitive pressure from digitally-enabled rivals serves as a major external catalyst, pushing SMEs to adopt cloud services to maintain market share and improve customer engagement (Jayeola et al., 2022:233; Abu Bakar et al., 2024:556). However, the intensity of this pressure is highly unequal: while urban SMEs face intense, immediate pressure from tech-dense competitors and digital platforms, rural enterprises operate in less competitive localized markets, reducing their perceived urgency to adopt cloud solutions (Sithole & Ruhode, 2021:296). Furthermore, government and vendor support represents a major institutional deficit in South Africa. While the DSBD has drafted policies aimed at accelerating SMME digitization, ground-level implementation and affordable technical training remain highly limited (DSBD, 2023:15). Local technology vendors also focus their support on large, high-revenue corporate clients, leaving small businesses with minimal guidance to navigate complex integration issues (Jayeola et al., 2022:237; Mokwena & Sithole, 2024:25).

Crucially, utility infrastructure constraints specifically persistent electricity supply disruptions (load shedding) and cellular connectivity gaps expose a fundamental difference between South African and global research (Adane, 2021:6; Mokwena & Sithole, 2024:28). Global TOE-based studies typically assume stable electricity and mature broadband networks as a given. In South Africa, however, load shedding frequently deactivates cellular base stations and local business routers, rendering cloud-hosted applications completely inaccessible (Adane, 2021:6). This critical vulnerability forces South African SMEs to maintain expensive hybrid or fully on-premises backups, thereby completely neutralizing the financial advantages of cloud configurations (Mokwena & Sithole, 2024:28; Adane, 2021:6).

3.6 Critical Synthesis of Literature

To systematically map the landscape of empirical research and identify theoretical points of convergence and divergence, Table 1 presents a comparative summary of key TOE-based cloud adoption studies.

Table 1: Critical Reviews on Related Studies of Cloud Adoption Studies

Author(s) & Year	Country / Scope	Methodology	Main Empirical Findings	TOE Dimension(s) Covered
Santos et al. (2023:Article 2220190)	Portugal	Quantitative Survey	Perceived relative advantage and technological compatibility are the strongest predictors of cloud adoption, which directly enhances MSME operational performance.	Technological, Organisational
Abu Bakar et al. (2024:556)	Malaysia	Quantitative Survey	Competitive pressure and active top management support represent the most influential drivers of cloud computing adoption intent among service-sector SMEs.	Organisational, Environmental
Yaseen et al. (2023:320)	Jordan	Quantitative Survey	Technological compatibility is the primary facilitator, whereas high architectural complexity acts as the dominant technological inhibitor to adoption.	Technological

Mokwena & Sithole (2024:18)	South Africa (Polokwane)	Quantitative Survey	High data costs, severe technical expertise shortages, and POPIA compliance anxieties act as severe barriers, resulting in low cloud adoption.	Technological, Environmental
Mkhize et al. (2025:14)	Sub-Saharan Africa	Systematic Review	Inadequate digital skills, a critical shortage of specialized IT personnel, and high mobile data costs represent the most significant barriers.	Organisational

3.8 Research Gap

A critical evaluation of Table 1 reveals several significant research gaps in the existing body of cloud computing adoption literature. Firstly, the vast majority of TOE-based studies have been conducted within highly stable developing or developed economies (such as Portugal, Malaysia, and Jordan), where a continuous national electricity supply and mature, well-supported legal compliance systems are assumed conditions that do not reflect the South African context (Adane, 2021:7; Mokwena & Sithole, 2024:30). Secondly, South African scholarship remains highly fragmented, frequently evaluating load shedding, POPIA compliance, and digital skills deficits in isolation rather than through an integrated, systematic theoretical lens (Adane, 2021:7). No evidence-based, contextualised cloud adoption framework specifically tailored to South Africa's unique regulatory, infrastructural, and economic conditions currently exists. This study directly addresses these gaps by applying a complete TOE framework to synthesize existing empirical research and propose a practical, context-sensitive adoption framework.

4. Design, Methodology and Ethics

4.1 Research Design

The research design selected for this study is a Systematic Literature Review (SLR) conducted in strict accordance with the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) statement (Page et al., 2021:Article n71). An SLR is the optimal design for resolving multi-dimensional adoption barriers by consolidating and critically synthesizing fragmented empirical findings into a unified theoretical framework. This design evaluates technology implementation not as a purely technical event, but as a complex sociotechnical process embedded within localized organisational limits and shifting institutional realities, such as electricity supply disruptions and POPIA regulatory constraints (Tornatzky & Fleischer, 1990:152; Mohlameane & Ruxwana, 2020:4). By adhering to a highly structured PRISMA-compliant selection protocol, the study eliminates selection bias, ensures transparency, and establishes a highly replicable research trail.

4.2 Research Methods and Data Sources

To gather a comprehensive, multi-disciplinary body of scholarly literature, the systematic search is executed across eight major academic databases: Scopus, Web of Science, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, Google Scholar, and MDPI (Mkhize et al., 2025:5; businesses-05-00023, 2025:8).

To isolate literature that specifically combines cloud configurations, SMEs, the TOE model, and the South African or developing-economy context, the database searches are executed using a precise Boolean search string across the title, abstract, and keyword fields:

("cloud computing adoption" OR "cloud adoption") AND ("SMEs" OR "small and medium enterprises" OR "SMMEs" OR "small businesses") AND ("Technology Organisation Environment" OR "TOE framework") AND ("South Africa" OR "developing economy" OR "developing nations").

Upon applying database-level filters restricting the results strictly to English-language publications, empirical articles or conference proceedings, and a temporal publication window of 2020 to 2026. This precise, recorded search trail will be executed across the remaining seven databases to establish a comprehensive search registry.

4.3 Study Selection and Inclusion/Exclusion Criteria

To filter the search results and ensure that only methodologically sound and highly relevant studies are included in the final review, a set of inclusion and exclusion criteria is established. Table 2 tabulates the exact operational boundaries applied during the screening process.

Table 2. Study inclusion and exclusion criteria.

Variable	Inclusion Criteria	Exclusion Criteria
Publication Window	Studies published between 2020 and 2026.	Studies published prior to 2020.
Language	English-language publications only.	All non-English language publications.
Document Type	Peer-reviewed journal articles, conference proceedings, and accredited master's or doctoral theses.	Editorials, blogs, book reviews, non-peer-reviewed white papers, opinion pieces, or reports.
Thematic Focus	Empirical studies evaluating cloud computing adoption specifically through the TOE framework.	Studies on unrelated sectors, non-cloud ICT architectures, or disconnected information systems models.
Geographic Scope	Research localized to South Africa or comparable developing and emerging economies.	High-income, highly developed Western or Asian market studies (e.g., USA, UK, Japan).

The screening workflow progresses systematically through three sequential phases. First, duplicates are automatically identified and removed across the databases. Second, the researcher conducts a title and abstract screen to eliminate papers that do not align with the core thematic focus or geographic scope. Third, full-text versions of the remaining papers are retrieved and evaluated in detail against the complete inclusion criteria to determine final eligibility. All exclusion decisions at this final stage are formally documented with justifications, as required by the PRISMA 2020 protocol (Page et al., 2021:Article n71).

4.4 Quality Appraisal

Every study meeting the inclusion criteria is subjected to a formal quality appraisal using the Mixed Methods Appraisal Tool (MMAT) Version 2018 (Hong et al., 2018:285). The MMAT is a standardized instrument designed to evaluate the methodological quality of qualitative, quantitative descriptive, quantitative non-randomized, and mixed-methods studies. The appraisal begins by evaluating two screening questions: (1) Are there clear research questions? and (2) Do the collected data allow for addressing the research questions? If a study satisfies both screening questions, it is evaluated against five design-specific criteria (e.g., for quantitative descriptive studies: sampling strategy appropriateness, sample representativeness, measurement validity, low risk of bias, and statistical analysis appropriateness) (Hong et al., 2018:286).

Each criteria is rated as 'Yes' (1 point), 'No' (0 points), or 'Can't tell' (0 points), resulting in an overall methodological quality score out of 5 (expressed as a percentage). Based on this score, studies are categorized into Quality Ratings: High Methodological Quality (80%-100%, 4-5 points), Moderate Methodological Quality (60%, 3 points), and Low Methodological Quality (<60%, under 3 points). To safeguard the scientific validity of the final framework, studies scoring under 60% are excluded from the review, ensuring that the theoretical synthesis is grounded exclusively in highly rigorous empirical evidence.

4.5 Data Extraction and Synthesis

To ensure a highly structured and transparent data collection process, a formal data extraction matrix is established.

Table 3. Outlines the precise Variable Data Collection parameters extracted from each eligible study.

Domain	Data Extraction Parameters	Socio-Technical Relevance Check
Bibliographic Data	Title, Lead Author, Year, Online Database, and Journal Name.	Tracks academic authority and publication recency (2020-2026 temporal boundary).

SME Study Context	Geographic Location, SME Industry Sector (manufacturing, ICT, finance), Sample Size, and Respondent Profiles.	Verifies developing-economy context and small business characteristics (under 200 employees).
Technical Model	Cloud Service Model (SaaS, PaaS, IaaS) and Cloud Deployment Model (Public, Private, Hybrid).	Identifies specific cloud configurations analyzed and maps them to SME requirements.
TOE Dimensions	Technological Context (advantage, complexity, security), Organisational Context (readiness, top management support), Environmental Context (rivals, support).	Classifies adoption drivers and barriers into Tornatzky and Fleischer's (1990) standard dimensions.
Socio-Technical Limits	Load Shedding Constraints, POPIA / Data Privacy compliance, cellular data costs, and Digital Literacy barriers.	Captures South Africa's unique infrastructural and regulatory hurdles to address the research gap.

The qualitative and quantitative findings extracted from the selected literature will be synthesized using Thematic Synthesis following the methodology of Thomas and Harden (2008). The synthesis progresses through three analytical stages: (1) the line-by-line coding of direct paragraph extractions, (2) the grouping of codes to develop descriptive themes, and (3) the generation of analytical themes. These analytical themes are mapped directly onto the Technological, Organisational, and Environmental dimensions of the TOE framework, enabling the researcher to build the contextualised cloud adoption framework for South African SMEs.

4.6 Ethics

Because this research is a systematic literature review utilizing exclusively published, secondary scientific sources, there are no risks to human participants. The researcher will submit a formal application to the CPUT Departmental Ethics Committee (under Dr. Kabaso) to secure a formal Ethical Clearance Exemption. To maintain academic integrity and prevent plagiarism, a strict referencing standard will be maintained, utilizing the CPUT Harvard style. The document will be analyzed using SafeAssign to ensure similarity indices remain well below CPUT's maximum similarity threshold of 15 percent.

In alignment with CPUT's institutional mandate to join the global 'open data' movement, the researcher commits to archiving all extracted data, screening sheets, and original PDF articles within a secure, highly organized local project directory (e.g., `Articles/ART-xxx\_Author\_Year.pdf`), which will be submitted on Blackboard upon final project delivery to ensure full auditability. Crucially, CPUT retains joint intellectual property ownership over the resultant adoption framework, which will be formally licensed under the open-source GNU General Public License (GPL) version 3.0 (Section F compliance) to ensure broad public accessibility and assist small business support agencies in modernizing SMMEs.

5. Delineation, Scope and Timeline

The scope of this study is strictly delineated to formal South African SMEs employing fewer than 200 individuals, in strict compliance with the thresholds established by the National Small Business Act (Act 102 of 1996, as amended) (DSBD, 2023:14). Geographically, the study focuses on factors relevant to the South African business environment, specifically local regulatory frameworks (POPIA) and infrastructural conditions (load shedding). Technologically, the review evaluates configurations across the three primary service models (SaaS, PaaS, IaaS) and deployment options (public, private, hybrid), using only the standard constructs of the TOE framework (relative advantage, compatibility, complexity, management support, readiness, competitive pressure, regulatory support, and infrastructure constraints).

To ensure the timely and successful completion of the research within the academic year, the project progresses according to a highly structured schedule.

Table 4. Presents the detailed Project Gantt Schedule and Milestone Breakdown.

Academic Phase	Core Activities & Operations	Target Completion Date	Deliverable & Output
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Phase 1: Proposal	Draft and submit research proposal, secure department ethical clearance exemption.	June 2026	Approved Proposal V3 & Exemption Form
Phase 2: Search	Execute Boolean search query across eight databases, remove duplicates, abstract screen.	August 2026	PRISMA Search Log & Screener Sheet
Phase 3: Appraisal	Retrieve full-text papers, evaluate study quality using MMAT checklist, extract data.	September 2026	Quality Appraisal Matrix
Phase 4: Synthesis	Conduct thematic analysis of extracted factors, design the contextualised TOE framework.	September 2026	Draft Cloud Adoption Framework
Phase 5: Reporting	Write final technical report, prepare academic poster, deliver oral presentation.	October 2026	Final Technical Report & Presentation

6. Significance

This study makes two highly interconnected contributions to theory and practice. Theoretically, it extends the application of the Technology-Organisation-Environment (TOE) framework into a unique, resource-constrained sub-Saharan African setting characterized by high utility volatility and a recently-enforced, complex data privacy regime (POPIA) (Tornatzky & Fleischer, 1990:152; Mohlameane & Ruxwana, 2020:4). While global TOE-based models treat basic infrastructure as a given and legal compliance as a minor administrative task, this research integrates utility failures (load shedding) and localized regulatory barriers directly into the theoretical model. This addresses the research gap by providing an integrated, socio-technical understanding of SME cloud adoption determinants (Adane, 2021:7).

Practically, the resulting cloud adoption framework serves as a strategic, evidence-based tool for multiple stakeholders. For SME owners and managers, the framework provides a structured roadmap to evaluate their organizational readiness, navigate technical complexities, and select legally compliant, affordable cloud service models (SaaS, PaaS, IaaS) (Mkhize et al., 2025:6). For government departments such as the Department of Small Business Development (DSBD), the framework identifies ground-level infrastructural and educational deficits that must be targeted through training programs and digital subsidies. Finally, for cloud service providers, the study offers actionable insights into the unique needs and barriers of local small businesses, enabling them to design highly tailored, affordable hybrid solutions that directly support the National Development Plan (NDP) goal of SME-led economic growth and poverty reduction (DSBD, 2023:12).

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